4.4.1 The Earth's atmosphere

The study of the development of the Earth's atmosphere shows how scientists base their theories on clues from the past that may be uncertain or incomplete.

Knowledge of the carbon cycle is crucial to understanding how human activities have changed the atmosphere on a global scale in ways that affect the climate.

Climate scientists explore climate change with the help of models. Earth systems are very complex and the data is often incomplete, so simplifying assumptions have to be made when setting up and testing the models that can then be used to evaluate possible methods for mitigating changes to the climate.

Human activities can also cause pollution on a more local scale, affecting air quality in areas with high traffic levels and contaminating water supplies with sewage.

Water cycles through the environment and is crucial to all living organisms. Various technologies have been developed to purify water so that it is safe to drink, and to treat sewage so that it does not harm the environment.

The required practical investigates the use of distillation to purify water.

4.4.1.1 Development of the Earth's atmosphere

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe how it is thought an oxygen-rich atmosphere developed over time.	Evidence for the early atmosphere is limited because of the time scale of 4.6 billion years. One theory suggests that during the first billion years of the Earth's existence there was intense volcanic activity, which released gases that formed the early atmosphere and water vapour that condensed to form the oceans. At the start of this period the Earth's atmosphere may have been like the atmospheres of Mars and Venus today, consisting mainly of carbon dioxide with little or no oxygen gas. Volcanoes also produced nitrogen, which gradually built up in the atmosphere, and there may have been small proportions of methane and ammonia. When the oceans formed, carbon dioxide dissolved in the water and carbonates were precipitated producing sediments, reducing the amount of carbon dioxide in the atmosphere. Algae and plants produced the oxygen that is now in the atmosphere by photosynthesis. Algae first produced oxygen about 2.7 billion years ago and soon after this oxygen appeared in the atmosphere. Over the next billion years plants evolved and the percentage of oxygen gradually increased to a level that enabled animals to evolve. Photosynthesis by algae and plants also decreased the percentage of carbon dioxide in the atmosphere. Carbon dioxide was also used up in the formation of sedimentary rocks, such as limestone, and fossil fuels such as coal, natural gas and oil.	WS 1.1 Given appropriate information, interpret evidence and evaluate different theories about the Earth's early atmosphere. WS 1.3 Explain why evidence is uncertain or incomplete in a complex context. MS 1c Use ratios, fractions and percentages.

4.4.1.2 The carbon cycle

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that many different materials cycle through the abiotic and biotic components of an ecosystem. Explain the importance of the carbon cycle to living organisms. Describe photosynthetic organisms as the main producers of food and therefore biomass for life on Earth. Explain the role of microorganisms in the cycling of materials through an ecosystem.	The element carbon is found as carbon dioxide in the atmosphere, dissolved in the water of the oceans, as calcium carbonate in sea shells, in fossil fuels and in limestone rocks, and as carbohydrates and other large molecules in all living organisms. Carbon cycles through the environment by processes that include photosynthesis, respiration, combustion of fuels and the industrial uses of limestone. Life depends on photosynthesis in producers such as green plants, which make carbohydrates from carbon dioxide in the air. Animals feed on plants, passing the carbon compounds along food chains. Animals and plants respire and release carbon dioxide back into the air. Decay of dead plants and animals by microorganisms returns carbon to the atmosphere as carbon dioxide and mineral ions to the soil.	WS 1.2 Draw and interpret diagrams to represent the main stores of carbon and the flows of carbon between them in the cycle. This topic links with Ecosystems and biodiversity (page 75).

4.4.1.3 The greenhouse effect

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the greenhouse effect in terms of the interaction of radiation with matter. (HT only) Recall that different substances may absorb, transmit or reflect these waves in ways that vary with wavelength.	Greenhouse gases in the atmosphere maintain temperatures on Earth high enough to support life. They allow short-wavelength radiation from the Sun to pass through the atmosphere to the Earth's surface but absorb the outgoing long-wavelength radiation from the Earth's surface, causing an increase in temperature. Water vapour, carbon dioxide and methane are greenhouse gases that increase the absorption of outgoing, long-wavelength radiation.	WS 1.2 Interpret and draw diagrams to describe the greenhouse effect.

4.4.1.4 Human impacts on the climate

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Evaluate the evidence for additional anthropogenic causes of climate change, including the correlation between change in atmospheric carbon dioxide concentration and the consumption of fossil fuels, and describe the uncertainties in the evidence base.	Human activities that involve burning fossil fuels (coal, oil and gas) for generating electricity, transport and industry all add carbon dioxide to the atmosphere. These activities have led to a large rise in the concentration of carbon dioxide in the air over the last 150 years. Over the same time the average temperature of the surface of the Earth has risen. The scientific consensus is that this is more than correlation and that the rise in greenhouse gas concentrations has caused the rise in temperature. Climate describes the long-term patterns of weather in different parts of the world. Climate change is shown by changes to patterns in measures of such things as air temperature, rainfall, sunshine and wind speed. Scientists analyse data on climate change using computer models based on the physics that describes the movements of mass and energy in the climate system. Many complex changes on Earth affect the climate, and detailed data about the scale of the changes is not available from all over the world. Also, when predicting climate change, scientists have to make assumptions about future greenhouse gas emissions. This means that there are uncertainties in the predictions.	WS 1.6 Explain the importance of scientists publishing their findings and theories so that they can be evaluated critically by other scientists. Understand that the scientific consensus about global warming and climate change is based on systematic reviews of thousands of peer reviewed publications. WS 1.3 Explain why evidence is uncertain or incomplete in a complex context. MS 2c, 4a Extract and interpret information from charts, graphs and tables. MS 2h Use orders of magnitude to evaluate the significance of data.

4.4.1.5 Climate change: impacts and mitigation

GCSE science subject content	**Details of the science content**	**Scientific, practical and mathematical skills**
Describe the potential effects of increased levels of carbon dioxide and methane on the Earth's climate and how these effects may be mitigated, including consideration of scale, risk and environmental implications.	Consequences of global warming and climate change include: • sea-level rise • loss of habitats • changes to weather extremes • changes in the amount, timing and distribution of rainfall • temperature and water stress for humans and wildlife • changes in the distribution of species • changes in the food-producing capacity of some regions. Steps can be taken to mitigate the effects of climate change by reducing the overall rate at which greenhouse gases are added to the atmosphere. Examples of mitigation include: • using energy resources more efficiently • using renewable sources of energy in place of fossil fuels (see Resources of materials and energy (page 141)) • reducing waste by recycling • stopping the destruction of forests • regenerating forests • developing techniques to capture and store carbon dioxide from power stations.	WS 1.4 In the context of climate change, evaluate associated economic and environmental implications; and make decisions based on the evidence and arguments.

4.4.1.6 Pollutants that affect air quality

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the major sources of carbon monoxide, sulfur dioxide, oxides of nitrogen and particulates in the atmosphere and explain the problems caused by increased amounts of these substances.	The combustion of fuels is a major source of atmospheric pollutants that can be harmful to health and the environment. Carbon monoxide is formed by the incomplete combustion of hydrocarbon fuels when there is not enough air. Carbon monoxide is a toxic gas that combines very strongly with haemoglobin in the blood. At low doses it puts a strain on the heart by reducing the capacity of the blood to carry oxygen. At high doses it kills. Sulfur dioxide is produced by burning fuels that contain some sulfur. These include coal in power stations and some diesel fuel burnt in ships and heavy vehicles. Sulfur dioxide turns to sulfuric acid in moist air. Oxides of nitrogen are produced by the reaction of nitrogen and oxygen from the air at the high temperatures involved when fuels are burned. Sulfur dioxide and oxides of nitrogen cause respiratory problems in humans and cause acid rain. Acid rain damages plants and buildings. It also harms living organisms in ponds, rivers and lakes. Particulates in the air include soot (carbon) from diesel engines and dust from roads and industry. The smaller particulates can go deep into people's lungs and cause damage that can lead to heart disease and lung cancer.	WS 1.4 Describe, explain or evaluate ways in which human activities affect the environment.

4.4.1.7 The water cycle

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the importance of the water cycle to living organisms.	Water is found in the solid state in glaciers and ice sheets, in the liquid state in the oceans, rivers, lakes and aquifers and in the gas state in the atmosphere. Water cycles through the environment by processes that include melting, freezing, evaporation and condensation. Precipitation of water from the atmosphere can take the form of rain, sleet or snow. Life on Earth depends on water, on land and in the seas. Water acts as the solvent for chemical reactions in cells. It also helps transport dissolved compounds into and out of cells. Water is either a reactant or a product of biochemical changes such as respiration, photosynthesis and digestion. Rivers, lakes and seas provide habitats for many living organisms.	WS 1.2 Draw and interpret diagrams to represent the main stores of water and the flows of water between them in the cycle.

4.4.1.8 Sources of potable water

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the principal methods for increasing the availability of potable water in terms of the separation techniques used, including ease of treatment of waste, ground and salt water. Describe, explain and exemplify the processes of simple distillation.	Water that is safe to drink is called potable water. Potable water is not pure water in the chemical sense because it contains dissolved substances. The methods used to produce potable water depend on available supplies of water and local conditions. In the UK, rain provides water with low levels of dissolved substances (fresh water) that collects in the ground and in lakes and rivers and most potable water is produced by: • choosing an appropriate source of fresh water • passing the water through filters • sterilising. Sterilising agents used for potable water include chlorine, ozone or ultraviolet light. If supplies of fresh water are limited, desalination of salty water or sea water may be required. Desalination can be done by distillation or by processes that use membranes such as reverse osmosis. Energy resources have to be used to run these processes. Urban lifestyles and industrial processes produce large amounts of waste water that require treatment before being released into the environment. Sewage and agricultural waste water require removal of organic matter and harmful microbes. Industrial waste water may require removal of organic matter and harmful chemicals. Sewage treatment includes: • screening and grit removal • sedimentation to produce sewage sludge and effluent • anaerobic digestion of sewage sludge • aerobic biological treatment of effluent.	WS 1.4 Explain everyday and technological applications of science; evaluate associated personal, social, economic and environmental implications with reference to the sources of potable water and treatment of waste water.

Required practical activity 11: analysis and purification of water samples from different sources, including pH, dissolved solids and distillation.

AT skills covered by this practical activity: chemistry AT 2, 3 and 4.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in Key opportunities for skills development (page 168).